

Online Exam Integrity Monitoring System

Project and Seminary

BSc in Computer and Software Engineering

Dinis Contreiras

A50496@alunos.isel.pt

934671411

Miguel Sanches

A49337@alunos.isel.pt

919772040

Supervisor:

Prof. Jorge Martins

jorge.martins@isel.pt

March 2026

Abstract

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1 Introduction

1.1 Outline

The widespread adoption of digital platforms in academic settings has transformed the way exams are performed, offering significant advantages when it comes to automation, accessibility, and even administrative efficiency for both educators and students. This transition comes with new challenges, particularly the preservation of academic integrity. When exams move from the paper world to laptops and computers, even with educators physically present in the room, the screen essentially becomes a private space for the student. This creates a meaningful gap where the professor is present but effectively blind to what is happening on each student's machine.

Online Exam Integrity Monitoring System (OEIMS) is a client-server monitoring system for controlled online exam environments.

On the client side, there's two main environments: student and professor. The student environment consists of a exam web application, accessed through a browser, and a lightweight background service running on the student's machine. The latter is responsible for both monitoring and mitigation during the exam. It collects integrity-related telemetry, including focus changes, suspicious process detection, network state changes, and heartbeat information, while also mitigating suspicious activity by blocking clipboard access and terminating forbidden processes. The professor environment consists of a web application that displays participant status and suspicious activity in real time.

In the other hand, the server side consists of a central server and a database. The system is built around a central backend server implemented with *Ktor* in Kotlin, which exposes a HTTP API used for authentication, exam management, session management and heartbeat updates. It uses *WebSocket* channels for real-time communication between the background service and the professor dashboard. The database is responsible for persisting the domain data containing users, sessions, participants and monitoring events but additionally, it stores session event history for posterior file-based logging.

1.2 Motivation

The motivation for this project emerged from a practical academic experience. During an *Erasmus* exchange period at Fontys University of Applied Sciences, in Venlo, the Netherlands, one of the authors observed that exams were commonly performed

using the student's own machine. The process was structured around a controlled digital environment, where students connected to the institutional network, executed the dedicated exam software, followed predefined instructions, and accessed the exam using a code provided shortly before it started. This experience showed how digital exams can be made practical for both students and teachers when the exam environment is properly designed and controlled.

This contrasted with the authors' experience in Portugal, where many exams are still on paper and where occasional digital exams, like in Computer Network course, may not provide the same level of control. In one observed digital exam context, students were technically able to access unauthorized resources, even if doing so was not allowed. This raised the idea of exploring how a digital exam environment could be supported by a monitoring system capable of detecting suspicious activity and that support supervisors.

2 Background

This chapter introduces the context needed to understand the problem addressed by this project, focusing on digital exams, exam supervision, and academic integrity.

Moving an exam from paper to a computer changes not only the format of the assessment, but also the environment that must be controlled. This chapter starts by presenting the difference between traditional paper exams and digital exams. It then introduces exam supervision as a long-established practice used to preserve trust and fairness in exams, before discussing how this changes when part of the student's activity happens inside the machine.

2.1 Digital Exams

A paper exam is an educational assessment with the traditional format where the statement, answers and submission are handled physically. The process depends entirely on physical material like paper for every document teachers might require. Normally, this would mean students receive printed exam sheets, write their answers on paper, and submit them to the supervisor at the end.

A digital exam follows the same academic purpose, but the exam process is performed through a computer, therefore, the student uses a machine to answer and submit the result. This changes the way the exam is organized, since the distribution, submission, storage and correction of the exam can be handled digitally [1, 2].

From an organizational point of view, digital exams can make the process more practical. Regarding the distribution, the process would no longer depend on estimating the number of students in advance, since the digital ecosystem can make the exam available independently of the number of participants. While in terms of correction, traditional exams depend on teachers to manually review each submission, which becomes time-consuming when there are hundreds of exams. Digital exams reduce this effort especially when the answers are suitable for automatic correction, allowing faster feedback to students [2]. Additionally, in some contexts, students would not have to pay for exam sheets beforehand, reducing paper usage and avoiding an extra cost that, although small, is repeated across courses and exam periods unnecessarily.

2.2 Exam Supervision

Examination supervision has a long history in education as a way to preserve trust in the assessment process. As written examinations became established in academic

institutions, especially in British universities during the nineteenth century, the role of institutional invigilators also appeared [3]. This shows that supervision is not a new concern created by digital exams.

Exam supervision, also called invigilation or proctoring in some contexts, is the process of observing and controlling the exam environment to help preserve exam integrity. Academic integrity is commonly associated with values such as honesty, trust, fairness, respect, responsibility, and courage [4]. In the context of an exam, supervision helps ensure that students perform the assessment under the expected conditions and do not use resources that are not allowed.

In paper exams, supervision is based on direct observation. The supervisor observes the room, the students, the desks, the materials and if suspicious behavior is identified, the supervisor can intervene because it happens in the same physical space.

In digital exams, supervision still has the same purpose, but the environment is different. The supervisor can still observe the classroom, but not everything that happens during the exam is necessarily visible from the outside since some activity takes place inside the student's machine, which makes the supervision process less dependent only on physical observation.

In this sense, supervision is part of the trust behind the assessment process: students are evaluated under the same conditions and teachers can have greater confidence that the work belongs to the respective student.

3 Problem Description

Although digital exams make the assessment process more practical, they also raise some challenges since they change the way supervision works. One important problem that arises is the lack of visibility over the computer environment. This is a problem because the student has at their disposal the operating system that offers numerous tools to break the exam rules without creating suspicion in the room.

Another challenge is the timing of the information. Suspicious activity is way more useful if it is detected during the exam, while the supervisor can still react. If this information is only discovered after the exam, it may still be useful for analysis, but the opportunity to intervene in the moment may already have been lost.

Reliability is also another challenge, especially because an exam can have a direct impact on a student's academic future and therefore does not allow much room for uncertainty. For instance, if failures and inconsistent states happen, it becomes harder for the institution to prove the supervision was trustworthy which affects the student unfairly. [3]

Another problem is misinterpretation. Accidents or unexpected behavior happen and every suspicious action should not represent academic misconduct. Some actions can happen accidentally or as part of the normal behavior of the operating system and, for this reason, supervision in digital exams must be understood with context.

Finally, modularity is another important challenge. The system should not be coupled to a single exam platform or way of conducting digital exams thus, it should be designed as a set of independent components that can be easily integrated. This makes the solution easier to maintain and more scalable for the future, although it is difficult to implement because it requires the problem to be decomposed into clear responsibilities, with well-defined interfaces between them, while avoiding dependencies that would make one part of the system aware of how another part works. [5].

Considering these challenges, we believe that a good system should address them through the following objectives:

- Monitor and deliver in real-time relevant activity inside each student's machine during the exam, including actions that may indicate the breakage of exam rules;
- Restrict predefined actions that are not allowed during the exam, reducing the dependence on later detection alone;
- Maintain a reliable and consistent representation of the exam state during the

exam session, so both supervisors and institution can trust and rely on the information;

- Provide a centralized, user-friendly platform that allows supervisors to manage the full lifecycle of exam sessions and access activity of each student's machine, cutting out the need to supervise each device manually or through separate tools;
- Preserve a record of relevant events of each student, allowing what happened afterwards to be reviewed with enough context to support fair and trustworthy decisions;
- Provide a modular and platform-independent structure that allows the system to be integrated with different ways of conducting digital exams, making it extensible.

3.1 Related Solutions

3.2 Proposed Solution

OEIMS (Online Exam Integrity Monitoring System) addresses this gap through four components: a background service running on each student's machine, a student exam web application, a central server, and a web-based professor console.

The two central components of this project are the background service and the professor console. The background service, referred to as the *daemon*, runs on the student's machine during the exam. It is responsible for monitoring operating system activity and reporting integrity-related events to the server as they occur, including focus changes, forbidden applications, clipboard usage, and network state deviations. Where appropriate, it also takes mitigating action to prevent access to resources not permitted during the exam. The professor console is a web application that provides the supervisor with a real-time view of all participants in a session, displaying the connection status and integrity events for each student as they are received.

The remaining components form the supporting infrastructure that connects these two. The central server manages authentication, session lifecycle, and event delivery, receiving events from all connected student machines, persisting them, and relaying them in real time to the professor console. The student exam web application provides the exam interface through which students access and submit their exam.

3.2.1 Functional Requirements

- Professors can create and manage exams, start sessions and generate a session code for students to join.
- Students authenticate with their institutional email and access the exam through a web application.
- The daemon detects suspicious behaviour including window focus loss, forbidden processes, clipboard usage and network changes.
- The daemon actively blocks clipboard access and attempts to terminate forbidden processes.
- A heartbeat mechanism detects client crashes or forced termination.
- Events are relayed in real time to the professor console, displaying each student's status and a timeline of occurrences.
- Session events are persistently logged on the server for later review.
- Multiple professors can supervise the same session simultaneously.
- Professors can join an active session using a session code.
- The system verifies that the daemon is connected before considering a student

ready in the session.

3.2.2 Non-Functional Requirements

- Session data and events transmitted between components should be secure, student authentication should be properly handled.
- The server should support multiple simultaneous exam sessions with a large number of concurrent students.
- The system should handle brief network interruptions gracefully without losing events.
- The project should be well documented to ensure maintainability and extensibility.

3.2.3 Optional Features

- Force full screen and block minimization during the exam.
- Block window switching.
- Block system shortcuts.

3.3 Limitations

It is important to note that the detection of such events does not automatically prove academic misconduct. A network disconnection may happen due to instability, a process may be launched unintentionally, and a window focus change may occur because of operating system behavior. Therefore, the role of the system is not to replace the professor's judgment, but to provide structured technical information that can back supervision during and after the exam.

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